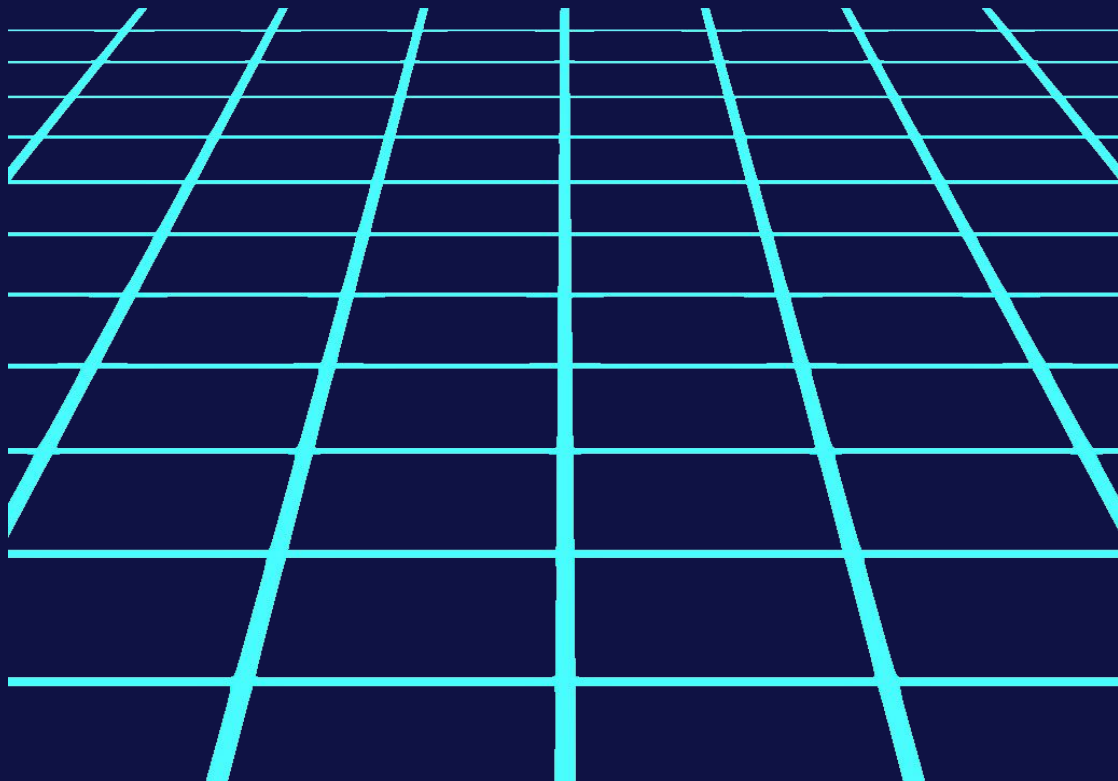




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Linux Cheat Sheet



File and Directory Management

1.ls - Lists directory contents.

- ls - Basic list.
- ls -l - List with details.
- Example: ls -la Directory

2.cd - Change the current directory.

- cd Directory - Changes to "Directory".
- Example: cd /home

3.pwd - Print the current directory path.

- pwd - Outputs the current directory.
- Example: pwd

4.mkdir - Create a new directory.

- mkdir Directory - Creates "Directory".
- Example: mkdir /home/Directory

5.rmdir - Remove an empty directory.

- rmdir Directory - Deletes "Directory".
- Example: rmdir /home/Directory

6. rm - Delete files and directories.

- `rm file` - Deletes "file".
- `rm -r Directory` - Deletes "Directory" recursively.
- Example: `rm -rf /home/Directory`

7. cp - Copy files or directories.

- `cp file new_file` - Copies "file" to "new_file".
- `cp -r Directory new_directory` - Copies "Directory" recursively.
- Example: `cp -r Directory /backup/`

8. mv - Move or rename files and directories.

- `mv file new_file` - Renames "file" to "new_file".
- Example: `mv file /home/Directory/`

9. touch - Create an empty file.

- `touch file` - Creates "file".
- Example: `touch /home/Directory/newfile`

10. cat - Concatenate and display file contents.

- `cat file` - Shows contents of "file".
- Example: `cat file`

11. less - View file content interactively.

- `less file` - Opens "file" in viewer.
- Example: `less file`

12. more - View file content page-by-page.

- `more file` - Views file with paging.
- Example: `more file`

13. head - Show the first few lines of a file.

- `head -n 5 file` - Shows first 5 lines.
- Example: `head file`

14. tail - Show the last few lines of a file.

- `tail -n 5 file` - Shows last 5 lines.
- `tail -f file` - Shows last lines with live updates.
- Example: `tail -f /var/log/syslog`

15. find - Search for files and directories.

- `find Directory -name "*.txt"` - Searches for .txt files.
- Example: `find /home -type f -name "*.sh"`

16. locate - Find files by name (needs mlocate package).

- `locate file` - Searches for "file".
- Example: `locate file`

17. du - Disk usage of files and directories.

- `du -sh Directory` - Shows size of "Directory".
- Example: `du -h Directory`

18. df - Display disk space usage.

- `df -h` - Disk usage in human-readable format.
- Example: `df -h`

19. stat - Display detailed file or directory status.

- `stat file` - Shows stats of "file".
- Example: `stat file`

20. tree - Show directory structure in tree format.

- `tree Directory` - Displays "Directory" structure.
- Example: `tree /home`

File Permissions and Ownership

21. chmod - Change file permissions.

- `chmod 755 file` - Sets permissions for "file".
- Example: `chmod +x script.sh`

22. chown - Change file ownership.

- `chown user file` - Changes owner to "user".
- Example: `chown root file`

23. chgrp - Change group ownership.

- `chgrp group file` - Changes group to "group".
- Example: `chgrp admin file`

24. umask - Set default permissions for new files.

- `umask 022` - Default file permissions.
- Example: `umask`

Process Management

- 25. ps** - Show active processes.
 - `ps aux` - Shows all processes.
 - Example: `ps aux | grep nginx`
- 26. top** - Monitor system processes in real time.
 - `top` - Live view of processes.
 - Example: `top`
- 27. htop** - Enhanced process viewer (requires `htop`).
 - `htop` - Interactive process viewer.
 - Example: `htop`
- 28. kill** - Terminate a process by ID.
 - `kill 1234` - Ends process with PID 1234.
 - Example: `kill 1234`
- 29. killall** - Kill processes by name.
 - `killall firefox` - Kills all "firefox" processes.
 - Example: `killall nginx`
- 30. pkill** - Kill processes by pattern.
 - `pkill -u user` - Kills all processes for "user".
 - Example: `pkill -u john`

Networking

31. ping - Check network connectivity.

- ping google.com - Sends ping requests.
- Example: ping -c 4 google.com

32. ifconfig - Display network configuration (deprecated).

- ifconfig - Shows network interfaces.
- Example: ifconfig

33. traceroute - Track packet route to destination.

- traceroute google.com - Shows route to "google.com".
- Example: traceroute google.com

34. nslookup - Query DNS records.

- nslookup google.com - DNS lookup for "google.com".
- Example: nslookup example.com

35. netstat - Network connections, routing tables.

- netstat -tuln - Lists open ports.
- Example: netstat -an

Compression and Archiving

37. tar - Archive multiple files.

- `tar -cvf archive.tar Directory` - Creates a tar archive.
- Example: `tar -xvf archive.tar`

38. zip - Compress files into a zip archive.

- `zip archive.zip file1 file2` - Creates "archive.zip".
- Example: `zip -r archive.zip Directory`

39. unzip - Extract files from a zip archive.

- `unzip archive.zip` - Extracts "archive.zip".
- Example: `unzip archive.zip`

40. gzip - Compress a file.

- `gzip file` - Compresses "file" as "file.gz".
- Example: `gzip -d file.gz`

41. bzip2 - Compress with bzip2.

- `bzip2 file` - Compresses "file".
- Example: `bzip2 -d file.bz2`

System Information

42. uname - Display system information.

- `uname -a` - Full system details.
- Example: `uname -r`

43. uptime - Show system uptime.

- `uptime` - Outputs how long system has been up.
- Example: `uptime`

44. free - Memory usage.

- `free -h` - Shows memory in human-readable form.
- Example: `free`

45. dmesg - Print or control kernel messages.

- `dmesg | less` - Shows boot logs.
- Example: `dmesg | grep error`

46. df - Disk space usage of filesystems.

- `df -h` - Human-readable format.
- Example: `df /home`

User Management

47. useradd - Create a new user.

- useradd username - Adds "username".
- Example: useradd admin

48. usermod - Modify user accounts.

- usermod -aG sudo user - Adds "user" to "sudo" group.
- Example: usermod -aG docker user

49. passwd - Change user password.

- passwd user - Changes password for "user".
- Example: passwd root

50. whoami - Display the current logged-in user.

- whoami - Shows username.
- Example: whoami

Data Manipulation

- 51. awk** - Pattern scanning and processing.
 - `awk '{print $1}' file` - Prints first column.
 - Example: `awk '{print $2}' file`
- 52. sed** - Stream editor for modifying files.
 - `sed 's/old/new/' file` - Replace text.
 - Example: `sed -i 's/foo/bar/g' file`
- 53. sort** - Sort file content.
 - `sort file` - Sorts file lines.
 - Example: `sort file`
- 54. uniq** - Report or omit repeated lines.
 - `uniq file` - Removes duplicates.
 - Example: `uniq -c file`
- 55. grep** - Search within files.
 - `grep "pattern" file` - Searches for "pattern".
 - Example: `grep -i error file`

File and Directory Management (Continued)

56. **wc** - Count lines, words, and characters in a file.

- **Usage:** `wc file` - Displays line, word, and character counts for "file".
- **Example:** `wc -l file` - Counts lines in "file".

57. **ln** - Create links between files.

- **Usage:** `ln file link_name` - Creates a hard link.
- `ln -s file link_name` - Creates a symbolic (soft) link.
- **Example:** `ln -s /path/to/file symlink_name`

58. **alias** - Create command shortcuts.

- **Usage:** `alias ll='ls -la'` - Creates an alias for `ls -la`.
- **Example:** `alias dir='ls -d */'`

59. **unalias** - Remove an alias.

- **Usage:** `unalias ll` - Removes the alias "ll".
- **Example:** `unalias dir`

60. **rename** - Rename multiple files.

- **Usage:** `rename 's/old/new/' *.txt` - Renames files by replacing "old" with "new" in .txt files.
- **Example:** `rename 's/2023/2024/' *.log`

Process Management (Continued)

61. bg - Run a job in the background.

- **Usage:** bg - Resumes a paused job in the background.
- **Example:** bg %1 - Backgrounds job number 1.

62. fg - Bring a job to the foreground.

- **Usage:** fg %1 - Brings job number 1 to the foreground.
- **Example:** fg %2

63. jobs - List all active jobs.

- **Usage:** jobs - Displays active jobs with their statuses.
- **Example:** jobs -l

64. nice - Run a command with adjusted priority.

- **Usage:** nice -n 10 command - Runs a command with priority 10.
- **Example:** nice -n 15 ./script.sh

65. renice - Change the priority of a running process.

- **Usage:** renice -n 5 -p PID - Changes the priority of process "PID".
- **Example:** renice -n 10 -p 1234

66. nohup - Run a command immune to hangups.

- **Usage:** nohup command & - Executes command, ignoring hangups.
- **Example:** nohup ./script.sh &

67. pgrep - Find processes by name.

- **Usage:** pgrep process_name - Finds process IDs with the specified name.
- **Example:** pgrep nginx

68. pmap - Display memory map of a process.

- **Usage:** pmap PID - Shows memory details of process "PID".
- **Example:** pmap 1234

69. uptime - Display system uptime and load.

- **Usage:** uptime - Shows system uptime and average load.
- **Example:** uptime

Networking (Continued)

70. curl - Transfer data to/from a server.

- **Usage:** curl URL - Fetches data from URL.
- **Example:** curl -O https://example.com/file.txt

71. wget - Download files from the internet.

- **Usage:** wget URL - Downloads file from the specified URL.
- **Example:** wget https://example.com/file.zip

72. scp - Securely copy files between hosts.

- **Usage:** scp source_file user@remote:/path - Copies file to a remote server.
- **Example:** scp file.txt user@remote-server:/home/user

73. rsync - Synchronize files and directories.

- **Usage:** rsync -av source/ destination/ - Syncs source to destination.
- **Example:** rsync -avz Directory user@remote:/backup/

74. telnet - Connect to remote servers (insecure).

- **Usage:** telnet host port - Connects to a specified host and port.
- **Example:** telnet localhost 80

75. netcat (nc) - Network utility for debugging.

- **Usage:** nc -zv host port - Checks if port is open on the host.
- **Example:** nc -zv google.com 443

76. ip route - Display or manipulate IP routing table.

- **Usage:** ip route show - Shows the routing table.
- **Example:** ip route add 192.168.1.0/24 dev eth0

77. dig - DNS lookup tool.

- **Usage:** dig domain.com - Fetches DNS records for a domain.
- **Example:** dig +short google.com

78. arp - Display or manipulate ARP cache.

- **Usage:** arp -a - Lists the ARP table.
- **Example:** arp -d IP_ADDRESS

Text Processing

79. echo - Display a line of text.

- **Usage:** echo "Hello World" - Outputs text to the screen.
- **Example:** echo \$PATH

80. tr - Translate characters.

- **Usage:** echo "text" | tr 'a-z' 'A-Z' - Converts lowercase to uppercase.
- **Example:** echo "hello" | tr 'a-z' 'A-Z'

81. cut - Remove sections from lines.

- **Usage:** cut -d ' ' -f 1 file - Displays the first field from each line.
- **Example:** cut -d: -f1 /etc/passwd

82. paste - Merge lines of files.

- **Usage:** paste file1 file2 - Joins lines from two files.
- **Example:** paste file1 file2

83. xargs - Execute commands from standard input.

- **Usage:** cat file | xargs echo - Passes arguments to echo.
- **Example:** find . -name "*.log" | xargs rm

84. nl - Number lines in a file.

- . **Usage:** nl file - Numbers each line.
- . **Example:** nl file

85. sort - Sort lines in a file.

- . **Usage:** sort file - Sorts lines alphabetically.
- . **Example:** sort -n file (numerical sort)

86. uniq - Filter out repeated lines.

- . **Usage:** uniq file - Removes duplicate lines.
- . **Example:** sort file | uniq

87. tee - Read from input and write to files.

- . **Usage:** command | tee file - Writes output to file and displays it.
- . **Example:** ls | tee file

Scheduling and Automation

88. cron - Schedule regular tasks.

- **Usage:** `crontab -e` - Opens the cron editor.
- **Example:** `0 5 * * * /path/to/script.sh` (Runs script daily at 5 AM)

89. at - Schedule one-time tasks.

- **Usage:** `echo "command" | at time` - Executes command at specified time.
- **Example:** `echo "backup.sh" | at 2:00 AM`

System Monitoring and Information

90. vmstat - Reports virtual memory statistics.

- . **Usage:** vmstat - Shows memory usage.
- . **Example:** vmstat 5 (updates every 5 seconds)

91. iostat - Reports CPU and I/O statistics.

- . **Usage:** iostat - Shows CPU and disk I/O stats.
- . **Example:** iostat -d 2

92. lsof - List open files.

- . **Usage:** lsof - Shows open files and processes.
- . **Example:** lsof | grep file

93. ss - Display socket statistics.

- . **Usage:** ss -tuln - Shows listening ports and connections.
- . **Example:** ss -s

94. free - Show memory usage.

- . **Usage:** free -h - Memory stats in human-readable format.
- . **Example:** free

95. mpstat - Display CPU usage.

- . **Usage:** mpstat - Shows per-processor CPU usage.
- . **Example:** mpstat -P ALL 1

96. sar - Collect and report system activity.

- **Usage:** `sar -u 1 5` - CPU usage every second for 5 times.
- **Example:** `sar -r` (memory stats)

Security and Access Control

97. iptables - Set up firewall rules.

- . **Usage:** iptables -L - Lists rules.
- . **Example:** iptables -A INPUT -p tcp --dport 22 -j ACCEPT

98. chmod - Change file permissions.

- . **Usage:** chmod 755 file - Sets permissions.
- . **Example:** chmod +x script.sh

99. chown - Change file ownership.

- . **Usage:** chown user:group file - Changes ownership.
- . **Example:** chown user file

100. passwd - Change user password. - **Usage:** passwd user - Sets a password for "user". - **Example:** passwd (for current user)

Thank
you



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